

CoRoT-2b

R-0221

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240807 · created 2026-07-17T01:37:41+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460529.7518 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.064
Transit depth (Rp/Rs)^2	4.01 +/- 2.58 [%]
Orbital Inclination (inc)	82.85 +/- 2.06
Airmass coefficient 1 (a1)	8.09 +/- 3.83
Airmass coefficient 2 (a2)	-1.56 +/- 0.71
Scatter in the residuals of the lightcurve fit is	57.45 %
Best Comparison Star	#3 - [624, 197]
Optimal Aperture	1.49
Optimal Annulus	6.44
Transit Duration (day)	0.066 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.2 $\pm$ 0.064 vs 0.1667 (deviation 0.52 σ)
Duration fit vs published	0.066 d vs 0.095 d (deviation 1.11 σ)
Mid-time O-C	-5.5 min
Depth SNR	3.1
Residual scatter	57.45 %

Light curve

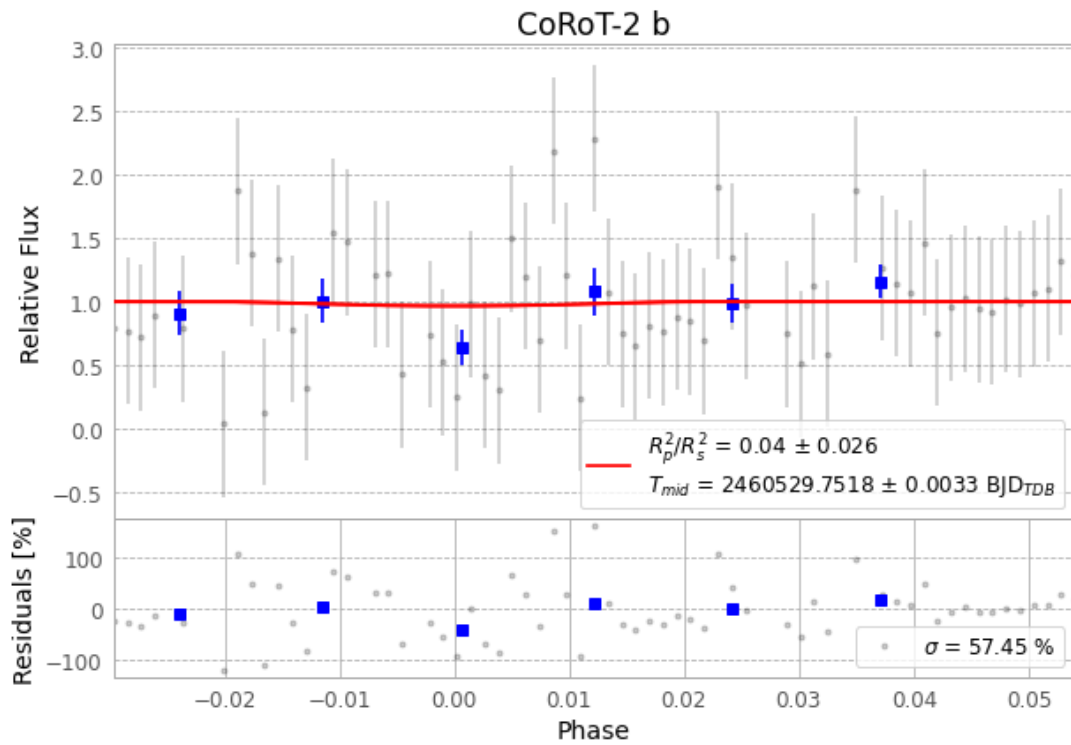


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-08-06.png (SHA-256 97bc1ffc628a5bf9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [282, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3f3111c853ca01ba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

71 FITS frames (47 MB), CoRoT-2240807043916.FITS → CoRoT-2240807080916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240807.log (e570ca8d3fcc...), FinalLightCurve_CoRoT-2 b_2024-08-06.png (97bc1ffc628a...), FinalLightCurve_CoRoT-2 b_2024-08-06.csv (b16251cbc923...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 01:37Z.